

Assignment

Module Code EM06DS
Module Name Big Data Analytics
Semester March 2026

Assessment Details and Submission Guidelines	
Assessment Type	Individual Project Portfolio (Hybrid: Engineering & Research)
Assessment Title	The Scalable Engineering & Research Portfolio: Digital Transformation in Aviation
Purpose of the Assessment (with CLO Mapping)	This assignment is designed to assess students' knowledge and skills related to the following learning outcomes: LO1 - Demonstrate deep understanding of the nature, evolution and relevance of big data and the impacts of big data on businesses and society. LO2 - Evaluate and analyse the advantages and limitations of big data and proposed practical solutions using appropriate big data processing frameworks. LO3 - Demonstrate sound knowledge of different data analytical techniques for different structured and unstructured big data sets to support decision-making. LO4 - Utilise contemporary libraries and tools to implement solutions for real-world challenges using neural networks and assess their performance rigorously. LO5 - Critically identify and select appropriate analytical technique for big data analysis using examples from case studies. LO6 - Critically evaluate and apply appropriate methods that are suitable for visualising big data.
Total Marks	100 marks
Word Limit	Research Paper (D2): 3,000 - 3500 words (Springer LNNS Template) Technical Report (D3): 1500 – 2000 words (OSEMN Framework)
Assessment Type	D1: Technical Artifact (50%) D2: Research Paper (30%) D3: Viva (20%)
Submission Guidelines	- A Comprehensive MS Word Report (.docx), a Research Paper (PDF), and a Python/PySpark code file (.ipynb or .py) must be submitted via Moodle. - Students must provide a GitHub/Databricks link within the report for code repository access.
Due Date	Students must submit the assignment by the end of Thursday, 10th Apr, 2026.
Late Submission	Late submissions will result in a penalty. A two-day late submission results in a 10% deduction on the project marking, while a submission between 2 and 4 days late will be subject to a 20% deduction on the project marking. Submissions received more than four days late will be considered a fail.
Academic Misconduct and Plagiarism	Plagiarism is defined as the presentation of another person's work as someone's own. This includes copying from books without referencing the material or copying from another student's work.
Module Leader	Dr. Zainab Rasheed
Moderator	Dr. Blessy Trencia Lincy Simon Shepherd

Individual Assignment Requirements

1. Project Overview

You are appointed as a **Senior Big Data Architect**. Your objective is to design and implement a distributed system capable of processing and analyzing an industry-scale dataset (Minimum **1 Million Records**). You must prove your system can handle the **Volume, Velocity, and Variety** of Big Data. This project requires moving beyond "laptop-scale" Python scripts and demonstrating mastery of distributed clusters, performance tuning, and empirical benchmarking.

2. Scalable Engineering: Detailed Task Descriptions (50 Marks)

These tasks comprise Deliverable 1. Evaluation is based on the technical complexity, scalability, and optimization of the code.

Phase A: Infrastructure & Distributed Preprocessing (25 Marks)

- **Task 1: Ingestion & Architectural Design (10 Marks | LO 1, 4)**
 - Activity: Architect a pipeline to ingest raw data into a distributed environment.
 - Technical Requirement: Explicitly define the Partitioning Strategy (e.g., partitioning by date or region) and justify cluster settings (Executor memory, Core allocation).
 - Tools: Hadoop HDFS, AWS S3 / Azure Blob Storage, Databricks File System (DBFS).
- **Task 2: Advanced ETL & Feature Engineering (15 Marks | LO 2, 4)**
 - Activity: Implement a distributed ETL pipeline to handle missing values, outliers, and normalization at scale.
 - Technical Requirement: Must use Lazy Evaluation and Distributed Transformations. Avoid any "collect()" operations that bring data to the local driver.
 - Tools: PySpark (DataFrames/SQL), Hadoop MapReduce (Java/Python).

Phase B: Scalable Deep Learning (10 Marks | LO 3)

- **Task 3: Distributed Deep Learning & Neural Networks.**
 - Activity: Develop a Neural Network architecture (e.g., Multi-Layer Perceptron) to address a high-volume aviation challenge. You must move beyond standard regression/classification and implement a multi-layered brain-inspired model.
 - Technical Requirement: Implementation of a Neural Network using contemporary libraries.
 - Rigorous Assessment: You must document the model's performance using a Confusion Matrix and Loss/Accuracy curves over multiple epochs.
 - Hyperparameter Tuning: Demonstrate the optimization of at least two parameters (e.g., learning rate, number of hidden layers, or batch size).
 - Tools: TensorFlow / Keras (via Google Colab) or PySpark MultilayerPerceptronClassifier.
 - **Evaluation:** Matplotlib/Seaborn (for Loss Curves), Scikit-Learn (for Confusion Matrix metrics).

Phase C: The Scalability Benchmark (10 Marks | LO 4, 6)

- **Task 4: Performance & Stress Testing**

- Activity: Run your pipeline under multiple configurations to test the limits of your architecture.
- Technical Requirement: Compare Execution Latency and Resource Consumption (CPU/RAM) across different data volumes (e.g., 20% vs 100%) or cluster sizes (e.g., 2 nodes vs 4 nodes).
- Tools: Spark UI (Environment/Storage tabs), Ganglia, Python (Matplotlib/Seaborn for benchmarking graphs).

Phase D: Visual Interpretation (5 Marks | LO 5)

- Task 5: Sophisticated Dashboarding
 - Activity: Design a dashboard that visualizes the "Insights" derived from the Big Data analysis.
 - Technical Requirement: Visuals must handle high-density data (e.g., Heatmaps, Trend projections) to assist executive decision-making.
 - Tools: Power BI, Tableau, Plotly Dash, or Streamlit.

3. Deliverable 2: Empirical Research Paper (30 Marks | LO 6)

Synthesize your engineering results into a **4,500-word** research paper formatted to the **Springer LNNS Template**.

- **Methodology:** Include a high-level **System Architecture Diagram**.
- **Engineering Results:** Present the **Benchmarking Graphs** (Task 4) and discuss "Speedup" and "Bottlenecks."
- **Ethics & Audit:** Address GDPR, data anonymization at scale, and algorithmic fairness.

4. Deliverable 3: Committee Viva & Report (20 Marks)

A defense of your work before a faculty committee.

- **PowerPoint Presentation:** Focus on the **Technical Logic** and **Scalability Proof**.
- **Comprehensive Report:** A single PDF containing the paper, architecture diagrams, and code snippets.
- **Viva Session:** Be prepared to justify your **Partitioning Strategy** and **Model Tuning** choices.

5. Assessment Criteria

Phase / Task	Title	Related LOs	Marks
Phase A / T1	Ingestion & Architecture	LO 1	10
Phase A / T2	Advanced ETL	LO 2	15
Phase B / T3	Distributed ML Pipeline	LO 3	10
Phase C / T4	Scalability Benchmarking	LO 4	10
Phase D / T5	Interactive Visualization	LO 5	5
Deliverable 2	Empirical Research Paper	LO 6	30
Deliverable 3	Committee Viva & Report	All LOs	20
TOTAL			100

6. Submission Checklist

- **D1: Technical Artifact:** GitHub link containing documented PySpark/Hadoop code and a README.md.
- **D1: Performance Log:** A PDF documenting the Spark UI logs and benchmarking graphs.
- **D1: Dashboard:** URL or exported file of the visual analytics.
- **D2: Research Paper:** PDF in Springer LNNS format.
- **D3: Viva Deck:** PowerPoint slides and the Comprehensive Final Report.

Appendix A: Research Paper Template

Format: Springer LNNS

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Author: [Student Name]

Affiliation: MSc in Artificial Intelligence and Data Science, Emirates Aviation University, UAE

Abstract

The abstract must be a single paragraph (200-250 words) structured as follows:

- **Background:** Brief context of the industry problem.
- **Problem:** Why do traditional "small-data" methods fail at this scale?
- **Methodology:** Mention the specific Big Data tools used (e.g., Spark MLlib, Hadoop HDFS).
- **Results:** Summarize the key findings (e.g., "The system achieved 92% accuracy with a 4x speedup on a 4-node cluster").
- **Conclusion:** The significance of the scalable solution.

1. Introduction

- **1.1 Motivation:** Discuss the importance of Big Data in your chosen domain (Aviation/Finance/Logistics).
- **1.2 Problem Statement:** Define the specific predictive or descriptive challenge.
- **1.3 Objectives:** List 3-4 clear goals (e.g., "To design a distributed ETL pipeline...").
- **1.4 Contributions:** State clearly what you have built (e.g., "This paper contributes a novel benchmarking analysis of Random Forest scalability on 10M records")

2. Literature Review

- Analyze 10-15 recent (past 5 years) Q1/Q2 journal papers.
- **Gap Analysis:** What is missing in current research that your "Scalable Engineering" approach addresses?
- *Note: Do not just list papers; synthesize them by theme.*

3. Methodology (The Scalable Architecture)

This is the most critical section for your Master's level marks.

- **3.1 System Architecture:** Provide a high-level diagram showing data flow from ingestion (HDFS/S3) $\rightarrow$ Preprocessing (Spark SQL) $\rightarrow$ ML Training (MLlib).
- **3.2 Dataset Description:** Source, volume (1M+ rows), features, and data types.
- **3.3 Data Partitioning & Shuffling:** Explain how you optimized the data for a distributed cluster. How did you handle the "Shuffle" phase?

4. Experimental Setup & Results

- **4.1 Hardware/Software Configuration:** Detail your cluster setup (e.g., "Databricks Runtime 13.3, 2-8 Worker Nodes, 14GB RAM each").
- **4.2 Analytical Results:** Present your ML metrics (Confusion Matrix, ROC Curve, AUC, F1-Score).
- **4.3 Engineering Results (Scalability Benchmark):**
 - **Scale-up:** Plot a graph showing "Execution Time vs. Data Size".
 - **Speed-up:** Plot a graph showing "Execution Time vs. Number of Nodes".

- **Discussion:** Why did the system scale this way? Were there bottlenecks?

5. Interpretation & Discussion

- Connect your results back to the industry. If you predicted flight delays, what do these results mean for an Airline Operations Manager?
- Compare your results with the works cited in Section 2.

6. Ethical, Legal, and Governance Audit

- **Data Privacy:** How did you ensure anonymity at scale?
- **Algorithmic Bias:** Did the model show bias towards certain demographics or categories in the large dataset?
- **Compliance:** Discuss how this architecture aligns with GDPR or relevant data sovereignty laws.

7. Conclusion & Future Work

- Summarize the technical and research success.
- Mention how this system could be scaled further (e.g., moving from Batch to Stream processing using Spark Streaming).

References

- Minimum 15 references.
- Use IEEE (Numerical) or Springer (Alphabetical) style consistently.
- Ensure at least 70% of references are from Q1/Q2 Journals or top-tier conferences (NeurIPS, ICML, KDD).

Note: To achieve the highest marks, your paper must not merely be a "lab report." It must:

1. **Justify the Framework:** Don't just say you used Spark; explain *why* Spark's in-memory processing was required for your specific dataset volume.
2. **Visual Excellence:** All charts must have labeled axes, legends, and high resolution (300 DPI).
3. **Critical Reflection:** Be honest about the limitations. If your scalability was sub-linear, explain why (e.g., network overhead or data skew).

Appendix B: Moodle Submission Requirements

Format Guidelines:

- **Font:** Times New Roman, Size 12 pt.
- **Spacing:** 1.5 line spacing.
- **Marginalia:** Standard 2.54 cm margins.
- **File Format:** All written reports must be submitted as **PDF** to preserve formatting, while code must be submitted in its native format.

1. Submission Bundle Checklist

Students must upload a single compressed folder (.zip or .rar) or individual files containing the following:

1. **Comprehensive Technical Report:** (Details below).
2. **Empirical Research Paper:** (As per the Springer Template in Appendix A).
3. **Python/Spark Source Code:** Jupyter Notebook (.ipynb) or Python script (.py).
4. **Viva Presentation:** PowerPoint slides (.pptx).
5. **External Link Document:** A one-page PDF containing the URL to your GitHub Repository or Databricks Public Archive.

2. Comprehensive Technical Report Structure

This report is the "Engineering Log" of your project. It should be structured as follows:

Front Matter

- **Cover Page:** Project Title, Student Name, ID, Module Code, and Date.
- **Table of Contents:** Automated with page numbers.
- **Abstract:** A 250-word summary of the technical achievement and performance results.

I. Introduction

- Scope of the project and selection of the Industry Domain.
- Definition of the "Big Data" nature of the chosen dataset.

II. Literature Review

- Brief synthesis of existing distributed solutions and tools related to your project theme.

III. Big Data Project Lifecycle (OSEMN Framework)

You must document your engineering process through the following six phases:

- **Phase 1: Obtain (Ingestion)**
 - Describe the data source and the ingestion method into HDFS/Databricks.
 - Explain the **Partitioning Strategy** used during ingestion.
- **Phase 2: Scrub (Scalable Preprocessing)**
 - Document how you handled missing values, outliers, and schema enforcement at scale.
 - Explain the use of **Distributed Transformations** (e.g., StringIndexer, VectorAssembler).
- **Phase 3: Exploratory Data Analysis (EDA)**
 - Visualizations showing distributions and correlations.
 - *Requirement:* Explain how you performed EDA on a massive dataset (e.g., using sampling or aggregate statistics).
- **Phase 4: Model (Distributed ML)**

- Details of the Spark MLlib pipeline, including hyperparameter tuning and cross-validation logic.
- **Phase 5: Interpret & Benchmark (Performance Analysis)**
 - Interpretation of the model results.
 - **Crucial:** Presentation of Scalability Graphs (Execution Time vs. Data Volume/Cluster Nodes).
- **Phase 6: Code Implementation & Snippets**
 - Provide screenshots of the most important parts of your Python/PySpark code.
 - **Annotation:** For every screenshot, provide a 2-3 sentence explanation of why that specific logic was used for a distributed environment.

IV. Conclusions

- Summary of engineering challenges overcome and final recommendations for the industry.

3. Python Code Submission Requirements

- **Format:** Jupyter Notebook (.ipynb) is highly preferred for its ability to show output and visualizations alongside code.
- **Environment:** If using Databricks, you must export your notebook as a "Databricks Archive (.dbc)" or a "Jupyter Notebook (.ipynb)".
- **Documentation:** Your code must be modular. Use Markdown cells to explain each stage of the Spark session and the ML pipeline.
- **Cleanliness:** Remove any failed execution attempts or unnecessary print statements before submission.

Appendix C: Viva Voce & Oral Defense Guidelines

1. Overview of the Viva Process

The Viva Voce (Oral Defense) is the final stage of your assessment. It is conducted by a committee of at least two faculty members. The purpose is to evaluate your depth of understanding, your technical proficiency in distributed environments, and the rigor of your research findings.

Duration: 20–30 Minutes

Format: 10-minute Presentation + 10-minute Technical Defense + Optional Live Code Stress Test.

2. Defense Structure

Part I: The Research Pitch (Presentation)

You will present your project using the PowerPoint slides submitted in Deliverable 3. Your presentation must focus on:

- **The Problem:** Why is this a "Big Data" problem? (Volume, Velocity, Variety).
- **The Architecture:** A high-level justification of your pipeline (e.g., Why did you choose Spark over MapReduce for this specific use case?).
- **The Breakthrough:** What did your scalability benchmarks reveal?
- **The Impact:** What is the industry value of your findings?

Part II: Technical & Methodological Defense

The committee will ask probing questions based on your Research Paper and Technical Artifact. Potential areas of questioning include:

- **Theoretical Grounding:** "Explain how the CAP theorem influenced your data storage strategy."
- **Logic Justification:** "Why did you choose a Hash Partitioning strategy over Range Partitioning for this dataset?"
- **Optimization:** "I see a significant shuffle read/write in your Spark UI logs. How would you refactor this code to minimize network overhead?"
- **Model Performance:** "How did you handle data skewness before training your MLlib model?"

Part III: The Code Stress Test (Live Modification)

- **Code Explanation:** You must explain the logic of a specific function or transformation in your notebook.

- **Small Modification:** You may be asked to change a parameter (e.g., the number of partitions or a hyperparameter in the ML pipeline) and explain what you expect the impact on performance to be.
- **Debug Thinking:** If a specific line of code is removed, what would be the impact on the distributed execution?

3. Preparation Checklist for Students

To succeed, you should be prepared to discuss the following:

1. **Comparative Analysis:** Be ready to explain why you didn't use a standard SQL database or a single-machine Python script.
2. **Data Integrity:** Explain how your "Scrub" phase handled the specific "Noise" of your industry dataset.
3. **Benchmarking Math:** Be able to explain the math behind your "Speedup" and "Efficiency" ratios.
4. **Ethical Awareness:** Defend your "Ethical Audit"—how would you prevent your model from being biased in a production environment?

4. Evaluation Criteria for the Committee

The committee will grade the viva based on:

- **Clarity of Communication:** Ability to explain complex distributed concepts simply.
- **Technical Authority:** Demonstrated mastery over the PySpark/Hadoop code and environment.
- **Critical Thinking:** Ability to justify engineering trade-offs (e.g., Accuracy vs. Latency).
- **Response to Challenge:** How the student handles technical "stress" and unexpected questions.

Appendix D: Grading Rubric

The following rubric provides detailed grading criteria for each component of the assignment.

Component	Distinction (70–100)	Merit (60–69)	Pass (50–59)	Fail (0–49)
1. Ingestion & Architecture (T1) (10 Marks)	Expert setup of Spark/Hadoop cluster. Sophisticated partitioning and shuffling strategy justified with architectural logic.	Good cluster configuration. Clear ingestion process with a basic partitioning strategy.	Adequate setup. Dataset ingested, but partitioning is either default or poorly justified.	Weak or non-distributed setup. Inappropriate for "Big Data" scale.
2. Advanced ETL at Scale (T2) (15 Marks)	Flawless implementation of distributed ETL. Uses lazy evaluation correctly; zero evidence of "local data collection" (no <code>.collect()</code>).	Strong ETL pipeline. Most transformations are distributed, with only minor gaps in optimization.	Basic cleaning and feature engineering. Some reliance on non-distributed or inefficient methods.	Failed to clean data at scale. Code is not modular or crashes on large volumes.
3. Distributed ML Pipeline (T3) (10 Marks)	Professional use of Spark MLib pipelines. Comprehensive hyperparameter tuning via distributed Grid Search/Cross-Validation.	Correct implementation of a Spark ML model. Standard tuning used with clear performance metrics.	Basic model implementation. Limited tuning or evaluation depth.	Incorrect model use or non-distributed implementation. No tuning evident.

Component	Distinction (70–100)	Merit (60–69)	Pass (50–59)	Fail (0–49)
4. Scalability Benchmarking (T4) (10 Marks)	Outstanding empirical analysis. Precise "Speedup" and "Latency" graphs comparing volume/nodes. Deep analysis of bottlenecks.	Clear performance logs provided. Good attempt at comparing configurations with accurate graphs.	Basic benchmarking. Mentions performance but lacks rigorous comparative data or clear visuals.	No benchmarking performed. Failure to prove system scalability.
5. Interactive Visualization (T5) (5 Marks)	Highly sophisticated dashboard (Power BI/Tableau/Plotly). High-density visuals that reveal complex big data insights.	Good quality dashboard. Visuals are clear and provide relevant industry insights.	Acceptable visuals, though may lack "drill-down" capability or handle large density poorly.	Poor or missing visualization. Charts do not offer meaningful data insights.
6. Empirical Research Paper (D2) (30 Marks)	Q1 Journal Standard. Exceptional synthesis of literature. Professional Springer/IEEE format. Clear contribution to IAMC-2026 themes. Deep Ethical/Legal audit.	High-quality academic writing. Strong methodology and results section. Minor formatting or citation gaps. Solid Ethical audit.	Descriptive paper. Meets basic academic standards but lacks critical depth in methodology or benchmarking discussion.	Poorly structured. Fails to meet 4500-word depth. Plagiarism or AI-dependency detected. No ethical discussion.

Component	Distinction (70–100)	Merit (60–69)	Pass (50–59)	Fail (0–49)
7. Committee Viva & Report (D3) (20 Marks)	Architectural Authority. Confidently defends choices. Handles code modifications with ease. Report is professional and uses OSEMN framework perfectly.	Good defense. Can explain main code components and justify most design choices. Report is well-organized.	Basic understanding. Struggles with deeper "Why" questions or code modification. Report has formatting issues.	Unable to explain own work. Failed the live code test. Report is missing major sections or OSEMN phases.

Appendix E: High-Impact Research Topics

Research Title	Data Type & Source	Methodology (Scalable Engineering)	Action Required (The "Gap")	Success Factors for Acceptance
Predictive Maintenance 4.0: A Distributed Deep Learning Approach for Cross-Fleet Engine Health Monitoring.	Time-Series Sensor Data (NASA CMAPSS or Airline Engine Logs).	Spark MLlib + LSTMs. Distributed training of Remaining Useful Life (RUL) models.	Compare "Local" vs "Distributed" training times.	Must include a Sensitivity Analysis of sensor noise at scale.
Urban Air Mobility (UAM) Traffic Flow: Scalable Conflict Detection Algorithms for High-Density Vertiport Operations.	Simulated Trajectory Data (BlueSky or OpenSky Network).	Graph Analytics (GraphX) to model aircraft as nodes in a dynamic spatial graph.	Benchmarking the system's ability to detect conflicts in <1ms.	Focus on Computational Latency as UAM requires real-time safety.
Green Aviation & Contrail Avoidance: Big Data Fusion of ADS-B and Satellite Meteorological Data for Fuel Optimization.	ADS-B Streams + NWP (National Weather Service) data.	PySpark SQL for massive spatial joins between flight paths and humidity layers.	Identify "Contrail-Sensitive" zones and calculate the fuel-burn trade-off.	Explicitly link results to Aviation Sustainability Goals (SDGs) .
Explainable AI (XAI) in ATM: Interpretable Machine Learning for Real-Time Air Traffic Controller Decision Support.	Historical Flight Tracks + Controller Logs.	Random Forests/XGBoost + SHAP/LIME values implemented in a distributed pipeline.	Move from "Black Box" predictions to "Reasoning" (e.g., Why was this delay predicted?).	Q1 journals love Human-in-the-loop validation (LO6).
Privacy-Preserving Passenger Analytics: A Federated Learning	CCTV/Biometric Metadata (Synthesized for privacy).	Federated Learning (Flower/Spark). Training	Prove that models can be trained without moving sensitive passenger data.	Address GDPR and Ethical Governance

Research Title	Data Type & Source	Methodology (Scalable Engineering)	Action Required (The "Gap")	Success Factors for Acceptance
Framework for Smart Airport Security Ingestion.		models across decentralized "Silos."		(Assignment Task 6).
Resilient Cargo Logistics: Big Data Analytics for Mitigating Global Supply Chain Disruptions in Air Freight.	IATA Cargo Volumes + Port Congestion Data.	Distributed Regression Analysis to find "Structural Breaks" in logistics flow.	Develop a "Resilience Index" for different airline cargo networks.	Provide a Comparative Study of pre- vs post-pandemic recovery.
Digital Twin Synchronization: Benchmarking Data Latency in Real-Time Aircraft Digital Twin Environments.	High-Frequency IoT Telemetry.	Spark Streaming (Structured Streaming) + Kafka integration.	Measure the "Sync-Gap" between the physical asset and the digital model.	Focus on the Engineering Bottlenecks (Assignment Phase C).
Aviation Revenue Management: Reinforcement Learning for Dynamic Pricing in High-Volatility Markets.	Global Distribution System (GDS) Price Scraping.	Distributed Reinforcement Learning (Ray/Spark).	Optimize seat pricing based on real-time competitor demand shifts.	Must discuss the Economic Implications (IAMC Topic).
Cyber-Physical Safety: Anomaly Detection in ADS-B Streams using Scalable Unsupervised Learning.	OpenSky Network (Raw ADS-B).	Isolation Forests / K-Means in Spark MLlib.	Detect "Ghost Aircraft" or signal spoofing in massive datasets.	High value for Aviation Security & Safety topics.
Human Factors & Pilot Fatigue: Analyzing Flight	Bio-sensor data (EEG/ECG) + Simulator Logs.	Feature Engineering on high-dimensional physiological data.	Correlate "Long-Haul" flight patterns with cognitive decline metrics.	Use Sophisticated Visuals (LO5) to

Research Title	Data Type & Source	Methodology (Scalable Engineering)	Action Required (The "Gap")	Success Factors for Acceptance
Deck IoT Data for Cognitive Load Prediction.				show stress heatmaps.
Biofuel Efficiency Analytics: Evaluating the Impact of Sustainable Aviation Fuels (SAF) on Engine Performance at Scale.	Engine Performance Logs (SAF vs Jet-A1).	Multivariate Analysis of Variance (MANOVA) on distributed clusters.	Quantify the exact efficiency loss/gain of SAF in various altitudes.	Direct alignment with IAMC Green Initiatives .
Disruption Management: A Scalable Heuristic for Real-Time Flight Rescheduling during Extreme Weather Events.	Historical Weather (NOAA) + Airport Delay Data.	Constraint Satisfaction Programming implemented in a Big Data framework.	Benchmark the time taken to re-route 1000+ flights simultaneously.	Focus on Operational Efficiency (IAMC Key Theme).
Aviation Tourism Economics: Sentiment Analysis of Social Media Streams for Predicting Post-Pandemic Travel Trends.	Twitter/X, TripAdvisor (API Scraped Data).	Distributed Natural Language Processing (NLP) using Spark NLP.	Extract "Travel Intent" and correlate with actual booking data.	Link Big Data to Aviation Finance & Economics .

END of ASSIGNMENT – GOOD LUCK.